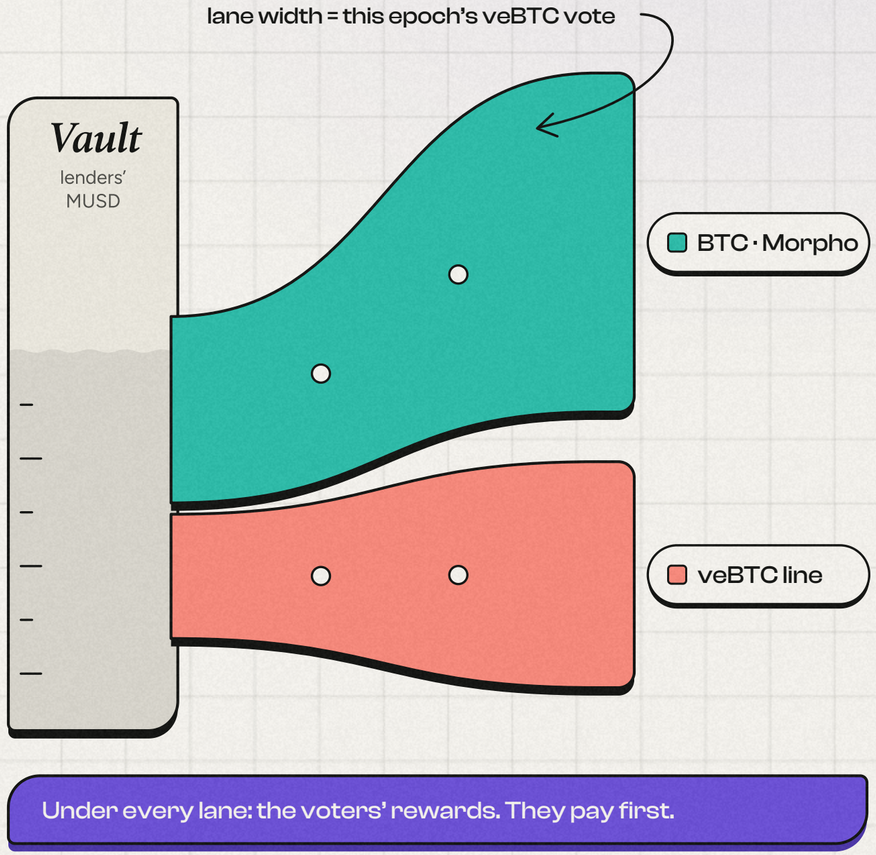
Surety

Lending gauges for Mezo

veBTC holders decide where MUSD credit goes, and their rewards take *the first loss*.

One MUSD vault, steered by veBTC votes. The voters' own rewards stand in front of every credit line they back.



Live on Mezo testnet

surety-credit.vercel.app

Yehor Ivashchenko

Surety

The problem

Plenty of BTC. Almost no *credit*.

804 BTC

locked in veBTC

Held by 1,796 holders. None of it can be borrowed against without unlocking. Mezo only does that for institutions, through Mezo Prime.

1

open trove

On Mezo's public MUSD CDP at block 12,705,315. Redemptions eat troves while MUSD trades about 1% under $1.

0

loss for whoever picks

In money markets today a curator or governance decides where deposits go. When the pick goes bad, it carries none of the loss.

Surety

From Mezo's blog · "Aerodrome for lending"

"...the system cannot tell who is serious"

mezo.org/blog/aerodrome-for-lending

WHAT MEZO ASKED FOR

Lending gauges: veBTC votes steer capital to lending markets, as a risk signal.

WHAT IT LEFT OPEN

The mechanism. It is saved for "a future post".

SURETY'S ANSWER

Make the vote cost something. The voters' rewards take the first loss.

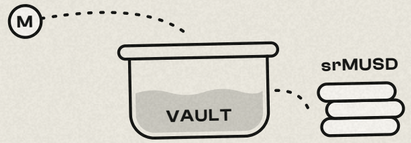
Surety

How it works

Four verbs, *one* vault.

Lenders bring the MUSD. veBTC holders steer it and stand behind their picks. Borrowers draw it with no redemption risk.

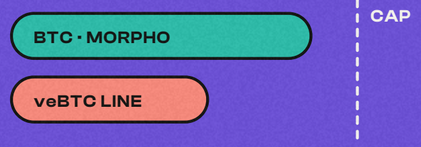
01 for lenders



Lend.

Deposit MUSD into a Morpho VaultV2 vault. You hold srMUSD shares.

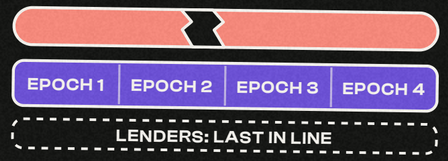
02 for veBTC holders



Vote.

veBTC power is snapshotted at epoch start. The allocator moves capital to match the votes, inside vault caps.

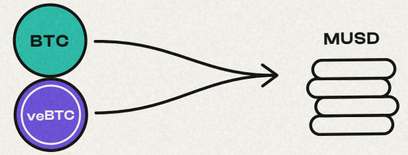
03 for the voters, again



Underwrite.

Voters earn 10% of interest plus sponsor incentives. Those rewards season 4 epochs and pay first if a line they backed goes bad.

04 for borrowers



Borrow.

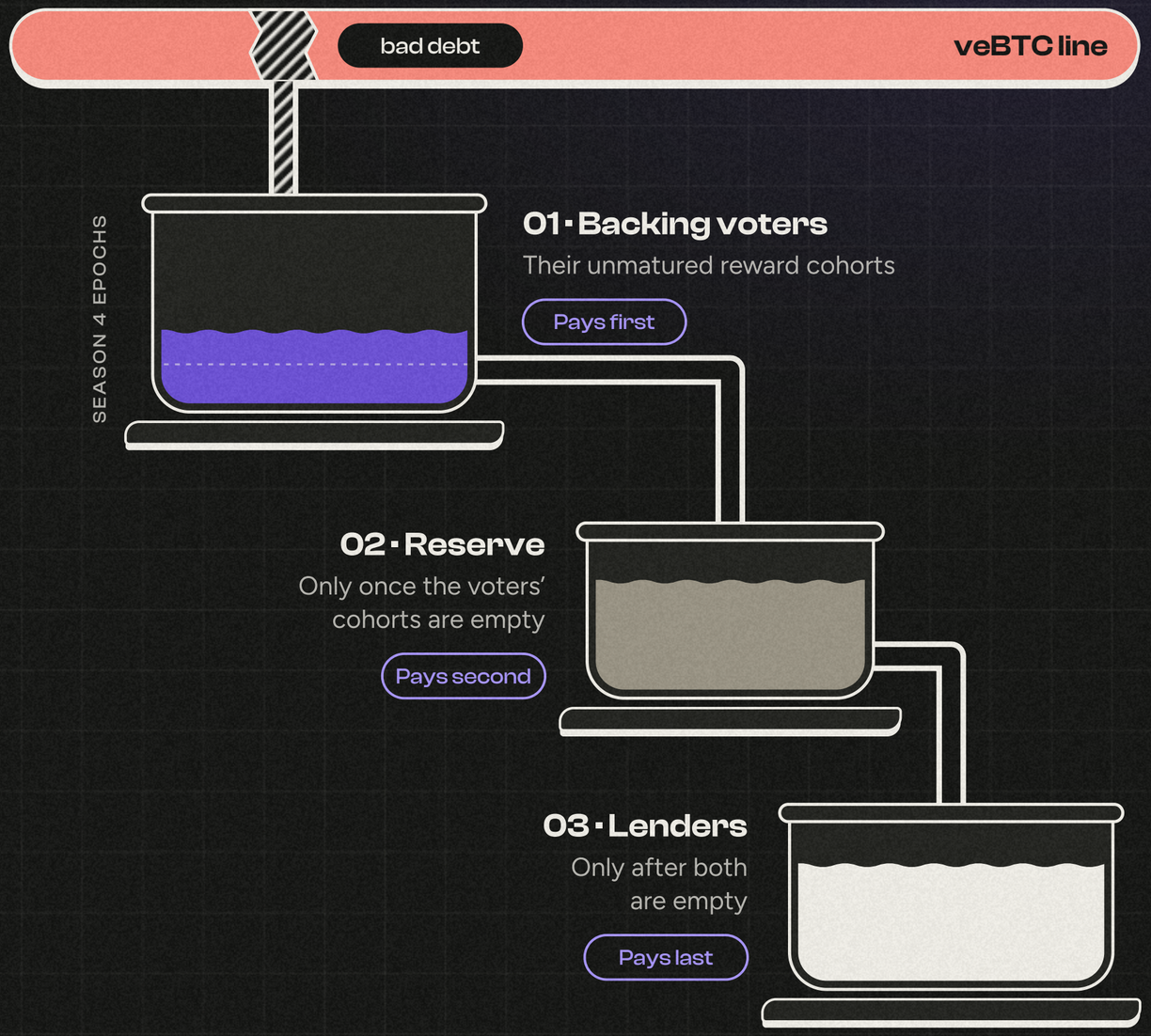
MUSD with no redemption risk: against BTC on Mezo's Morpho Blue, or against a locked veBTC that keeps voting.

Surety

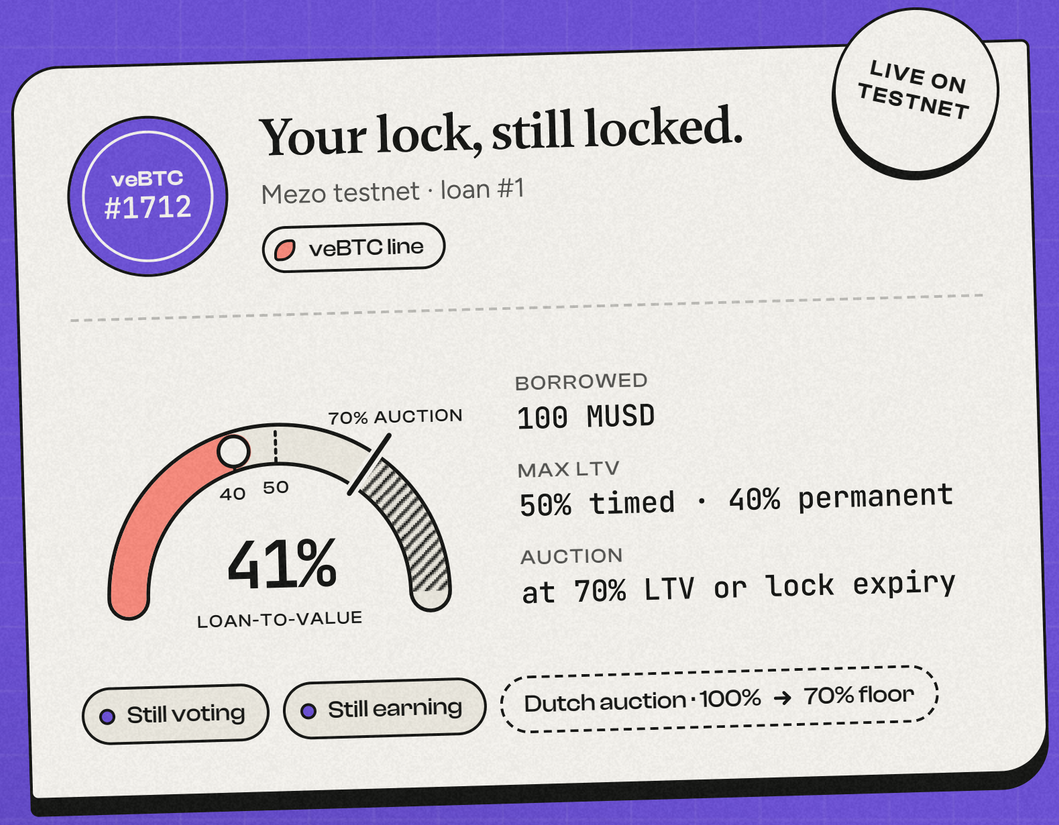
Who pays first

When a loan goes bad, the voters pay first.

Order	Backing voters' unmatured cohorts, then the reserve, then lenders.
Size	Cover is sized so lenders return to the pre-loss share price.
Gate	New deposits pause while a cover is re-recognised, so nobody can skim it.



Surety



Mezo Prime does this for institutions. Surety does it for any lock.

For veBTC holders

Borrow against your veBTC without unlocking it.

Stays locked	Each loan gets its own escrow. The lock keeps voting and earning on Mezo.
Borrow up to	50% LTV on a timed lock, 40% on a permanent one.
If it slips	At 70% LTV or at lock expiry, a Dutch auction runs from 100% of the lock's value down to a 70% floor.
No redemptions	The MUSD comes from the vault, so nobody can redeem against you.

Surety

Built on Mezo

Made from Mezo's *own* parts.

MUSD 01

`lent · borrowed · incentives`

What lenders deposit, borrowers draw and sponsors post.

veBTC 02

`votingPowerOfNFTAt`

Vote weight, snapshotted at the start of each epoch.

Mezo voting + rewards 03

`PoolsVoter · RewardsDistributor`

Reached through each loan's escrow, so a pledged lock keeps voting and claiming.

Skip BTC/USD 04

`precompile · fail-closed at 60 s`

Prices every loan and auction. A price older than 60 s stops them.

Morpho Blue on Mezo 05

`LLTV 0.86 · AdaptiveCurveIrm`

The BTC market the vault supplies, where BTC holders borrow MUSD.

VaultV2 + adapter 06

`MorphoMarketV1AdapterV2`

Morpho's vault and adapter, vendored unmodified at the exact commits Mezo runs.

GAUGE-READY

Mezo's own `ERC4626VaultAdapter` fits the vault. One governance call switches on MEZO emissions for lenders.

Surety

Live on Mezo testnet

Current deployment · all 14 contracts Sourcify exact match · every step is a transaction

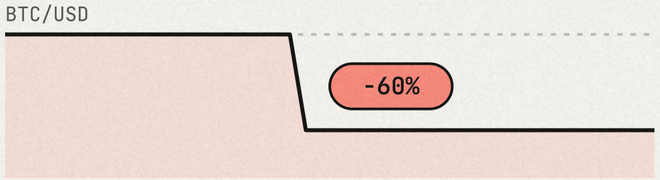
One run, from deposit to bad debt.

SETUP — lend, vote, borrow

01	Deposit 0xb0f5…1b7f	1,000 MUSD
02	Vote · veBTC #1711 0x457d…3c19	70 / 30
03	Vote · veBTC #1712 0xa237…2168	30 / 70
04	Rebalance 0xea9d…d272	450 / 450 MUSD
05	Sponsor incentives 0x4028…0bd9 · 0xc9ad…a124	100 + 150 MUSD
06	veBTC loan 0x050d…9fe9	100 MUSD · 41% LTV
07	Morpho borrow 0xb8b2…b9d2	250 MUSD · 76.9% LTV

STRESS — break it on purpose

08	BTC price shock 0x1a75…0434	-60%
09	Morpho liquidation 0x9fd3…0723	125.40 bad debt
10	veBTC auction 0x32c4…79ad	68.24 vs debt 100



veBTC shortfall 31.76 MUSD

COVER — voters first

11	Record loss · Morpho 0x34c0…00a8	
12	Record loss · veBTC 0x9aeb…da00	
13	Cover · Morpho line 0x7a52…8b03	
14	Cover · veBTC line 0xf260…ae9b	

Deposits paused
automatically while the covers re-recognise

Surety

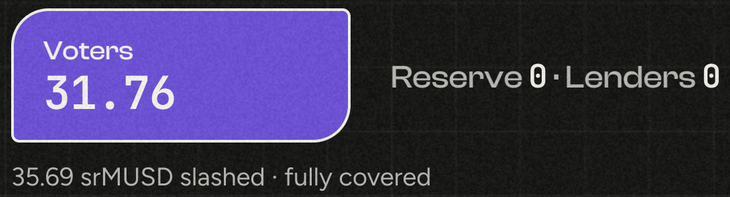
Who paid · same testnet run

157.16 MUSD of bad debt. Lenders bore *36.98*.

veBTC line

31.76 MUSD

Auction at the floor: 68.24 bid vs 100 debt

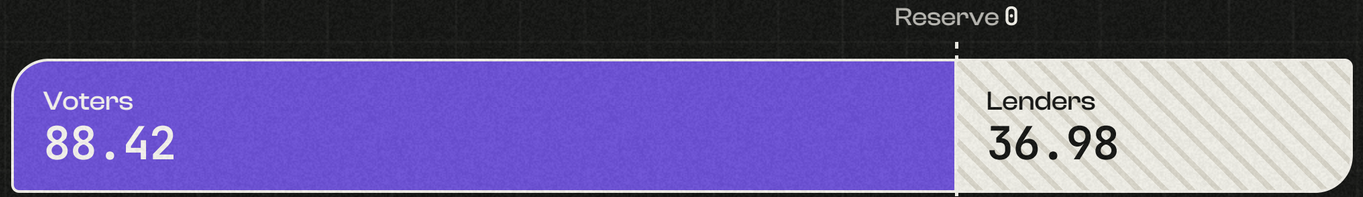


35.69 srMUSD slashed · fully covered

BTC · Morpho

125.40 MUSD

Liquidation left bad debt



Cohort of 100 srMUSD used up

What the cohort could not cover

DEPOSIT GATE New deposits paused automatically during re-recognition, so nobody could buy in cheap and skim the cover.

Surety

Engineering & security

Tested like it already holds *money*.

197

Foundry tests

Including 7 invariants at 256 runs × 64 depth.

9

regression tests

Written from an independent security review.

0

critical or high

The review found 4 medium and 6 low. All fixed.

99

TypeScript tests

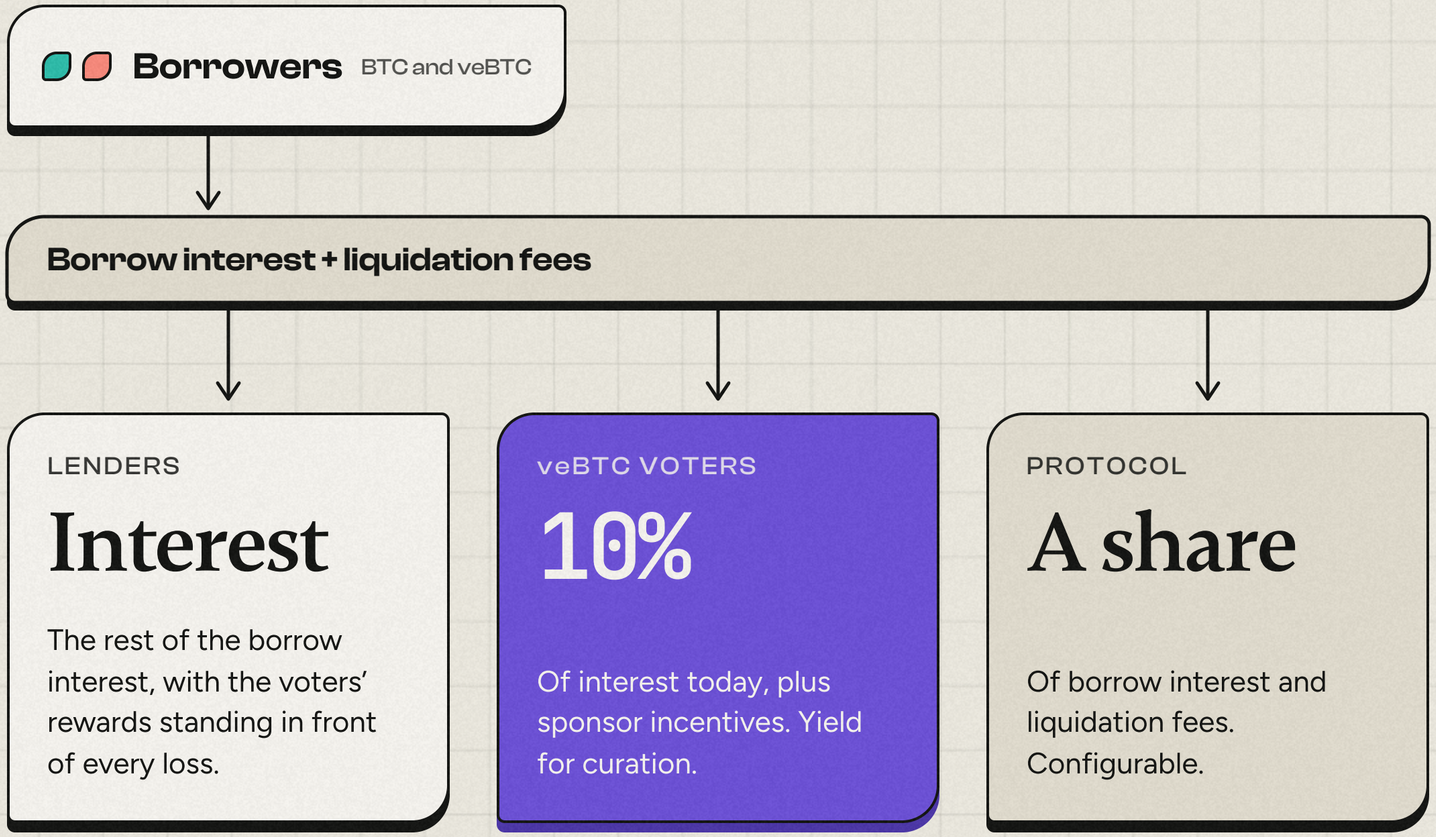
Across the SDK, the keeper and the MCP server.

SHIPS WITH

- Permissionless keeper
- TypeScript SDK
- MCP server for agents
- Web app · `surety-credit.vercel.app`

Business

Borrowers pay for the whole loop.



WHAT MEZO GETS

01 Demand for MUSD.

02 A use for veBTC beyond fees.

03 The lending-gauge mechanism its blog asked for.

Surety

Roadmap

Capped first. Then *open it up*.

NOW

Live on Mezo testnet, 14 contracts verified

Live on Mezo mainnet, capped

NOV 2026 · WAVE 2

Real Mezo vault gauge, through governance

MEZO incentives

Self-repaying veBTC loans

DEC 2026

Owner roles to timelock + multisig

External audit

Q1 2027

Raise caps

More BTC collateral lines

Surety

Try it

Votes with something *on the line*.

LIVE APP

surety-credit.vercel.app

REPOSITORY

github.com/RaYYeR220/surety

VIDEO

youtu.be/dHxYqFmqjEg

MAINNET

Live, capped · vault 0x632d...45de

BUILT BY

Yehor Ivashchenko solo developer

X

@BazydloMis57623

Surety